



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	QUANTUM MECHANICS II				
Type of Course	Core in Major				
Semester	VII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. Fundamental Physical and Mathematics Concepts of Quantum Mechanics				
Course Summary	Delves deeper into the mathematical formalism and theoretical principles of quantum theory, exploring topics such as advanced wave function theory, scattering theory, perturbation theory, etc.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Demonstrate Proficiency in Solving Schrödinger Equation Problems in Spherical Polar Coordinates.	U, Ap	C, P	Instructor-created exams / Quiz
CO2	Analyze Angular Momentum Concepts and Apply Them to Quantum Systems.	An, Ap	C, P	Practical Assignment / Observation of Practical Skills
CO3	Construct and Interpret Eigenvalues and Eigenfunctions of Angular Momentum Operators	C, U	C, P	Seminar Presentation / Group Tutorial Work

CO4	Evaluate Perturbation Theory Techniques for Solving Quantum Mechanical Problems.	E	C, P	Instructor-created exams / Home Assignments
CO5	Critically Analyze Scattering Phenomena and Predict Experimental Outcomes.	An, E	C	One Minute Reflection Writing assignments
CO6	Synthesize Advanced Quantum Mechanical Concepts to Solve Complex Problems.	C	C, M	Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	THE CENTRAL POTENTIAL		12	14
	1	Schrodinger Equation in Spherical Polar Coordinates : Separation of variables, The angular equation and spherical harmonics.	5	
	2	The Radial Equation,	1	
	3	Infinite Spherical Well,	2	
	4	The Hydrogen Atom : Radial Wave Function, Spectrum of Hydrogen	4	
	Sections: 4.1, 4.2 and Example 4.1 of Book 1			
II	ANGULAR MOMENTUM		13	20
	5	Angular Momentum : Orbital angular momentum, General formalism of angular momentum	3	

	6	Matrix representation of angular momentum, Geometrical representation of angular momentum.	1	
	7	Eigenfunctions of Angular Momentum - L_z only.	1	
	8	Spin : Experimental evidence of spin, General theory of spin, spin $\frac{1}{2}$ and Pauli matrices.	3	
	9	Addition of Angular Momenta : Addition of Two Angular Momenta General Formalism, Calculation of the Clebsch–Gordan Coefficients.	5	
Sections: 5.1 – 5.7.1 and 7.3.1 – 7.3.2 of Book 2				
III	APPROXIMATION METHODS		14	24
	10	Non-degenerate Perturbation Theory: First-order and Second-order theory	2	
	11	Degenerate Perturbation Theory : Two-fold degeneracy, Higher-order degeneracy.	1	
	12	Zeeman effect: Weak-field Zeeman effect, Strong-field Zeeman effect	1	
	13	Intermediate field Zeeman effect	2	
	14	Stark Effect	1	
	15	The Variational Method: Theory	1	
	16	Example: 1D Harmonic Oscillator	1	
	17	WKB Approximation: WKB wavefunction in classical and non-classical (tunneling) region, Connection Formula.	3	
	18	Examples: Potential well with one vertical wall, Potential well with no vertical walls	2	
	Sections: 6.1, 6.2, 6.4, 7.1, Example - 7.1, 8.1, 8.2, 8.3, Examples - 8.3 and 8.4 of Book1			

IV	SCATTERING		9	12
	19	Classical Scattering Theory	1	
	20	Quantum Scattering Theory	1	
	21	Partial Wave Analysis: Formalism, Strategy, Phase Shifts	3	
	22	Born Approximation: Integral Form of Schrödinger Equation, First Born Approximation, Born Series.	4	
	Sections: 11.1, 11.2, 11.3, 11.4 of Book 1			
*Solved and unsolved problems of the relevant sections from the prescribed texts shall be discussed or given as assignment.				
V	PRACTICALS		30	
	Conduct any 6 experiments from the given list and 1 additional experiment, decided by the teacher-in-charge, related to the content of the course. The 7 th experiment may also be selected from the given list. Other experiments listed here may be used as demonstrations of the concepts taught in the course.			
	1	Photoelectric effect: Determination of Plank’s constant.		
	2	Frank Hertz experiment: To measure the ionization potential of Mercury by drawing current versus applied voltage.		
	3	Elementary experiments using Laser: (a) Study of Gaussian nature of laser beam (b) Evaluation of beam spot size (c) Measurement of divergence (d) Diameter of a thinwire.		
	4	Zeeman effect using Fabry-Perot etalon.		
	5	ESR spectrometer - Determination of g factor		
	6	Thomson’s e/m measurement - To determine charge to mass ratio of the electron by Thomson’s method.		
	7	Millikan’s oil drop experiment - To measure the charge on the electron.		
	8	Simulate the Particle in a one dimensional box		

9	Simulate the Particle in a quadratic potential.		
10	Simulate the Quantum mechanical Tunnel barrier problem- Study the variation of transmission probability with L, E, V, and m.		
11	Simulate the Hydrogen wave functions (s, p, d,f) using 3D plots.		
12	Simulate the formation of wave packets as function of number of mixing waves		

Books and References:

1. Introduction to Quantum Mechanics, David J Griffiths, 3ed Edition. (Book 1)
2. Quantum Mechanics Concepts and Applications, Nouredine Zettili, 2nd Edition (Book 2)
3. Modern Quantum Mechanics by J. J. Sakurai and Jim Napolitano
4. Principles of Quantum Mechanics by R. Shankar
5. Quantum Mechanics: A Modern Development by Leslie E. Ballentine
6. Quantum Mechanics: Non-Relativistic Theory by L. D. Landau and E. M. Lifshitz.
7. NPTEL video lectures: <https://nptel.ac.in/courses/122106034>
8. Python for Education by Dr. B P Ajithkumar, IUAC, New Delhi; e-book freely downloadable from <https://scischool.in/python/index.html>

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO 4	PSO 5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	0	2	0	3	1	2	0	1	1	3	1	0
CO 2	3	2	3	0	3	2	2	0	1	1	3	1	0
CO 3	3	1	3	0	3	2	2	0	1	1	3	1	0
CO 4	3	0	3	0	2	1	3	0	1	1	3	1	0
CO 5	3	1	3	0	2	1	3	1	1	1	3	1	0
CO 6	3	0	2	0	3	1	3	0	1	1	3	1	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	